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DATA-260 Section 22
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Configuration:
SID4: 4544
PORT_BASE: 8044
PREFIX: s4544
SEED: 4544
VERIFY_SEED: 264544
DOMAIN_ID: 4 (Open_source package vulnerabilities)

Hardware: Apple M3, 16GB
Model: qwen2.5:3b (switched from qwen3.8b due to hardware issues)

Recent repo state (w/o report): e201bcb25689d2ded25d632ea8896cef1fc3b747
Link: https://github.com/sangahlee0/data260-4544
Link to HW6 Commit#:
https://github.com/sangahlee0/data260-4544/tree/e201bcb25689d2ded25d632ea8896cef1fc3b747 (This commit link does not include the report, because it will be added after downloading this document)

Homework 6

PART 1 - MongoDB Document API

I. Database

- Create MongoDB database “s4544_doc”

config.py

```python
class Settings(BaseSettings):
    MONGODB_URI: str = "mongodb://localhost:27017"
    MONGODB_DB: str = "s4544_doc"
```

```
test> use s4544_doc
switched to db s4544_doc
s4544_doc> show collections
```

- Include at least: one required short text field (max 100 characters), one optional description field, two constrained categorical fields (e.g., a status-style enum with a default, and a category-style field with a fixed set of options), one required date/time field, and automatic created/updated timestamps
- Schema must include proper validation rules such as required fields, string length limits, and enum constraints

schemas.py

```python
class VulnerabilityCreate(BaseModel):
    vulnerability_name: str = Field(min_length=1, max_length=100)
    description: Optional[str] = None
    status: Literal["open", "investigating", "resolved"] = "open"
    severity: Literal["low", "medium", "high", "critical"]
    discovered_at: datetime


class VulnerabilityUpdate(BaseModel):
    vulnerability_name: Optional[str] = Field(
        default=None,
        min_length=1,
        max_length=100
    )
    description: Optional[str] = None
    status: Optional[Literal["open", "investigating", "resolved"]] = None
    severity: Optional[Literal["low", "medium", "high", "critical"]] = None
    discovered_at: Optional[datetime] = None


class VulnerabilityOut(BaseModel):
    id: str
    vulnerability_name: str
    description: Optional[str] = None
    status: str
    severity: str
    discovered_at: datetime
    created_at: datetime
    updated_at: datetime
```

II. API Implementation

- Implement a REST API using FastAPI with complete CRUD operations

POST - vulnerabilities.py

```python
@router.post("", response_model=VulnerabilityOut, status_code=status.HTTP_201_CREATED)
def create_vulnerability(payload: VulnerabilityCreate):
    col = get_vulnerabilities_collection()
    doc = payload.model_dump()

    ts = now_utc()
    doc["created_at"] = ts
    doc["updated_at"] = ts

    try:
        res = col.insert_one(doc)
        created = col.find_one({"_id": res.inserted_id})
        return serialize_vulnerability(created)
    except PyMongoError:
        raise HTTPException(
            status_code=500,
            detail="Database error while creating vulnerability."
        )
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8044/api
/vulnerabilities -H "Content-Type: application/json" -d '{"vulnerability_name":"lodash prototype pollution","severity":"
high","discovered_at":"2026-10-16T00:00:00"}'
{"id":"6ad42035ed97e50c60c73fc0","vulnerability_name":"lodash prototype pollution","description":null,"status":"open","s
everity":"high","discovered_at":"2026-10-16T00:00:00","created_at":"2026-10-18T01:26:13.683000","updated_at":"2026-10-18
T01:26:13.683000"}
HTTP 201
```

GET - vulnerabilities.py

```python
@router.get("", response_model=list[VulnerabilityOut])
def get_all_vulnerabilities():
    col = get_vulnerabilities_collection()

    try:
        docs = list(col.find().sort("created_at", -1))
        return [serialize_vulnerability(d) for d in docs]
    except PyMongoError:
        raise HTTPException(
            status_code=500,
            detail="Database error while fetching vulnerabilities."
        )
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8044/api/vulnera
bilities
[{"id":"6ad42035ed97e50c60c73fc0","vulnerability_name":"lodash prototype pollution","description":null,"status":"open","
severity":"high","discovered_at":"2026-10-16T00:00:00","created_at":"2026-10-18T01:26:13.683000","updated_at":"2026-10-1
8T01:26:13.683000"},{"id":"6aced8835d8387f0fde16076","vulnerability_name":"SQL Injection","description":"Injection vulne
rability found in login endpoint","status":"open","severity":"high","discovered_at":"2026-10-13T18:00:00","created_at":"
2026-10-14T01:18:59.377000","updated_at":"2026-10-14T01:18:59.377000"},{"id":"6aced76e5d8387f0fde16075","vulnerability_n
ame":"string","description":"string","status":"open","severity":"low","discovered_at":"2026-10-14T01:14:18.949000","crea
ted_at":"2026-10-14T01:14:22.855000","updated_at":"2026-10-14T01:14:22.855000"},{"id":"6aced76c5d8387f0fde16074","vulner
ability_name":"string","description":"string","status":"open","severity":"low","discovered_at":"2026-10-14T01:14:18.9490
00","created_at":"2026-10-14T01:14:20.629000","updated_at":"2026-10-14T01:14:20.629000"}]
HTTP 200
(venv) (base) sangah@MacBook-Air-SAL data260-4544 %
```

GET by id - vulnerabilities.py

```python
@router.get("/{vulnerability_id}", response_model=VulnerabilityOut)
def get_one_vulnerability(vulnerability_id: str):
    if not is_valid_object_id(vulnerability_id):
        raise HTTPException(
            status_code=400,
            detail="Invalid id format."
        )

    col = get_vulnerabilities_collection()

    try:
        doc = col.find_one({
            "_id": to_object_id(vulnerability_id)
        })

        if not doc:
            raise HTTPException(
                status_code=404,
                detail="Vulnerability not found."
            )

        return serialize_vulnerability(doc)

    except HTTPException:
        raise
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8044/api/vulnera
bilities/6ad42035ed97e50c60c73fc0
{"id":"6ad42035ed97e50c60c73fc0","vulnerability_name":"lodash prototype pollution","description":null,"status":"open","s
everity":"high","discovered_at":"2026-10-16T00:00:00","created_at":"2026-10-18T01:26:13.683000","updated_at":"2026-10-18
T01:26:13.683000"}
HTTP 200
(venv) (base) sangah@MacBook-Air-SAL data260-4544 %
```

PUT - vulnerabilities.py

```python
@router.put("/{vulnerability_id}", response_model=VulnerabilityOut
def update_vulnerability(
    vulnerability_id: str,
    payload: VulnerabilityUpdate
):
    if not is_valid_object_id(vulnerability_id):
        raise HTTPException(
            status_code=400,
            detail="Invalid id format."
        )

    col = get_vulnerabilities_collection()

    try:
        existing = col.find_one({
            "_id": to_object_id(vulnerability_id)
        })

        if not existing:
            raise HTTPException(
                status_code=404,
                detail="Vulnerability not found."
            )

        updates = payload.model_dump(exclude_unset=True)
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % curl -s -w "\nHTTP %{http_code}\n" -X PUT http://127.0.0.1:8044/api/
vulnerabilities/6ad42035ed97e50c60c73fc0 -H "Content-Type: application/json" -d '{"status":"resolved"}'
{"id":"6ad42035ed97e50c60c73fc0","vulnerability_name":"lodash prototype pollution","description":null,"status":"resolved
","severity":"high","discovered_at":"2026-10-16T00:00:00","created_at":"2026-10-18T01:26:13.683000","updated_at":"2026-1
0-18T01:29:59.809000"}
HTTP 200
```

DELETE - vulnerabilities.py

```python
@router.delete(
    "/{vulnerability_id}",
    status_code=status.HTTP_204_NO_CONTENT
)
def delete_vulnerability(vulnerability_id: str):
    if not is_valid_object_id(vulnerability_id):
        raise HTTPException(
            status_code=400,
            detail="Invalid id format."
        )

    col = get_vulnerabilities_collection()

    try:
        res = col.delete_one({
            "_id": to_object_id(vulnerability_id)
        })

        if res.deleted_count == 0:
            raise HTTPException(
                status_code=404,
                detail="Vulnerability not found."
            )

        return

    except HTTPException:
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % curl -s -w "\nHTTP %{http_code}\n" -X DELETE http://127.0.0.1:8044/a
pi/vulnerabilities/6ad42035ed97e50c60c73fc0

HTTP 204
(venv) (base) sangah@MacBook-Air-SAL data260-4544 %
```

- Include proper error handling with appropriate HTTP status codes for scenarios like invalid data, resource not found, and server errors

```python
if res.deleted_count == 0:
    raise HTTPException(
        status_code=404,
        detail="Vulnerability not found."
    )
```

```python
except PyMongoError:
    raise HTTPException(
        status_code=500,
        detail="Database error while deleting vulnerability."
    )
```

```python
if not updates:
    raise HTTPException(
        status_code=400,
        detail="No fields provided to update."
    )
```

```python
    raise HTTPException(
        status_code=400,
        detail="Invalid id format."
    )
```

- Use middleware for CORS and JSON parsing for request bodies. Structure your code with separate files for server configuration, PyMongo database connection, data models/schemas (Pydantic), and route handlers

main.py

```python
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from routers.vulnerabilities import router as vulnerabilities_router
from routers.memory import router as memory_router

app = FastAPI(title="Vulnerability API (FastAPI + PyMongo)")

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=False,
    allow_methods=["*"],
    allow_headers=["*"],
)

app.include_router(vulnerabilities_router)
app.include_router(memory_router)

@app.get("/")
def health():
    return {"status": "ok", "message": "Study Planner API is running"}
```

III. Data Validation

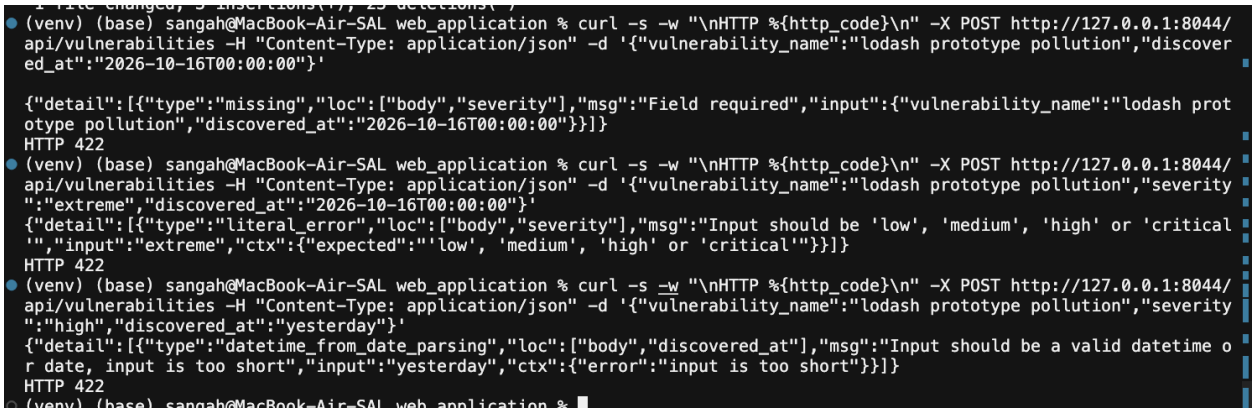
- Implement validation using Pydantic schemas to enforce data integrity including required fields, string lengths, enum values, and proper date formats.

schema.py

```python
class VulnerabilityCreate(BaseModel):
    vulnerability_name: str = Field(min_length=1, max_length=100)
    description: Optional[str] = None
    status: Literal["open", "investigating", "resolved"] = "open"
    severity: Literal["low", "medium", "high", "critical"]
    discovered_at: datetime
```

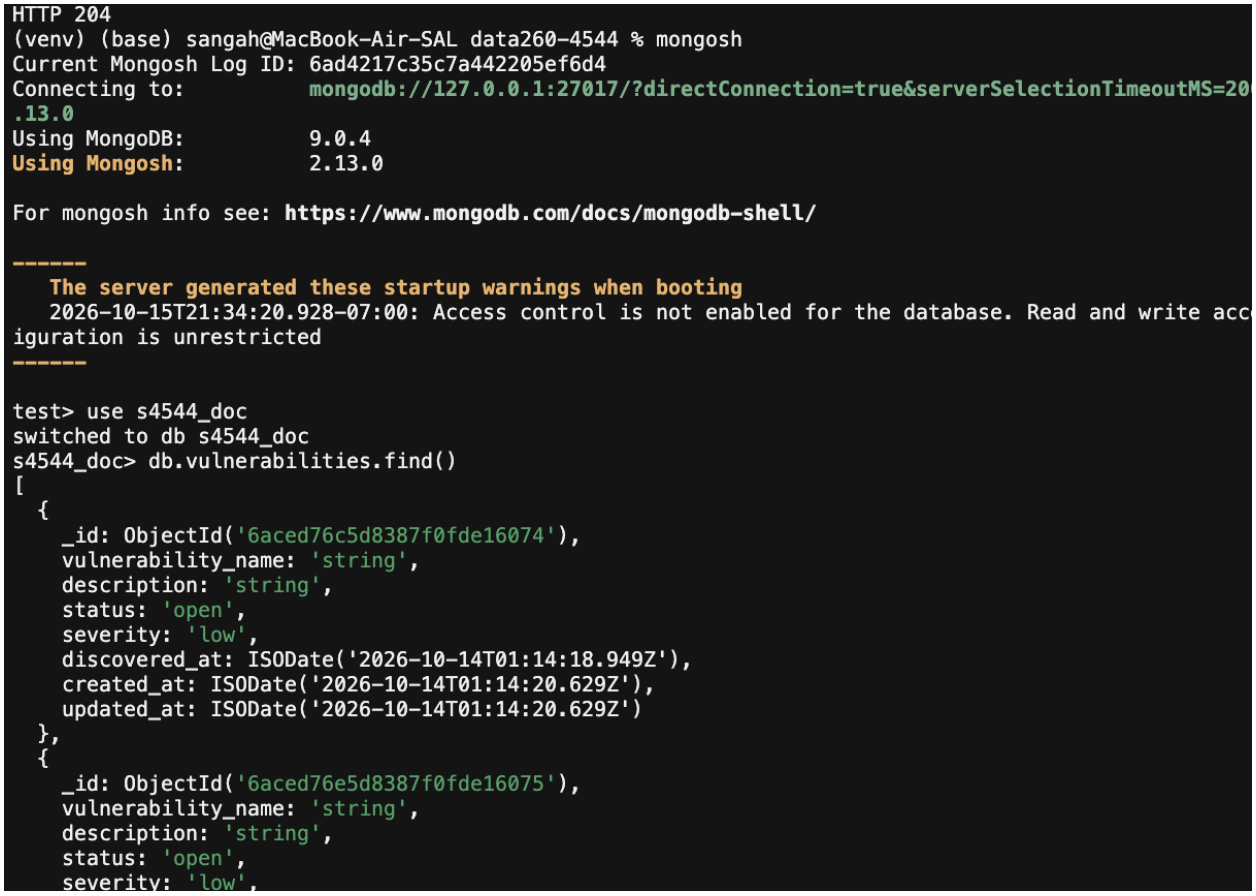
- Ensure that validation errors return clear error messages with appropriate HTTP status codes

- Test validation by attempting to create records with invalid data such as missing required fields, invalid enum values, and incorrect data types



```
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8044/
api/vulnerabilities -H "Content-Type: application/json" -d '{"vulnerability_name":"lodash prototype pollution","discover
ed_at":"2026-10-16T00:00:00"}'

{"detail":[{"type":"missing","loc":["body","severity"],"msg":"Field required","input":{"vulnerability_name":"lodash prot
otype pollution","discovered_at":"2026-10-16T00:00:00"}}]}
HTTP 422
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8044/
api/vulnerabilities -H "Content-Type: application/json" -d '{"vulnerability_name":"lodash prototype pollution","severity
":"extreme","discovered_at":"2026-10-16T00:00:00"}'
{"detail":[{"type":"literal_error","loc":["body","severity"],"msg":"Input should be 'low', 'medium', 'high' or 'critical
'","input":"extreme","ctx":{"expected":"'low', 'medium', 'high' or 'critical'"}}]}
HTTP 422
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8044/
api/vulnerabilities -H "Content-Type: application/json" -d '{"vulnerability_name":"lodash prototype pollution","severity
":"high","discovered_at":"yesterday"}'
{"detail":[{"type":"datetime_from_date_parsing","loc":["body","discovered_at"],"msg":"Input should be a valid datetime o
r date, input is too short","input":"yesterday","ctx":{"error":"input is too short"}}]}
HTTP 422
(venv) (base) sangah@MacBook-Air-SAL web_application %
```



```
HTTP 204
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % mongosh
Current Mongosh Log ID: 6ad4217c35c7a442205ef6d4
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=20
.13.0
Using MongoDB:          9.0.4
Using Mongosh:          2.13.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-10-15T21:34:20.928-07:00: Access control is not enabled for the database. Read and write acc
iguration is unrestricted
------

test> use s4544_doc
switched to db s4544_doc
s4544_doc> db.vulnerabilities.find()
[
  {
    _id: ObjectId('6aced76c5d8387f0fde16074'),
    vulnerability_name: 'string',
    description: 'string',
    status: 'open',
    severity: 'low',
    discovered_at: ISODate('2026-10-14T01:14:18.949Z'),
    created_at: ISODate('2026-10-14T01:14:20.629Z'),
    updated_at: ISODate('2026-10-14T01:14:20.629Z')
  },
  {
    _id: ObjectId('6aced76e5d8387f0fde16075'),
    vulnerability_name: 'string',
    description: 'string',
    status: 'open',
    severity: 'low',
```

PART 2 - Short-, Long-, and Episodic Memory

I. FastAPI Endpoints (memory.py)

1. POST /api/chat

```python
@router.post("/api/chat")
def chat(payload: ChatRequest):
    if payload.session_id:
        session_id = payload.session_id
    else:
        session_id = "default"

    try:
        # save user message
        save_message(payload.user_id, session_id, "user", payload.message)

        # save up to 3 facts from that message
        facts = extract_facts(payload.message)
        for fact, importance in facts:
            embedding = embed_text(fact)
            save_episode(payload.user_id, session_id, fact, importance, embedding)
```

Response:

```python
return {
    "reply": reply,
    "session_id": session_id,
    "memory_used": {
        "short_term_count": len(recent_messages),
        "short_term_messages": short_term_messages,
        "long_term_summary": long_term_summary,
        "episodic_facts": episodic_facts,
    },
```

2. GET /api/memory/{user_id}

```python
@router.get("/api/memory/{user_id}")
def get_memory(user_id: str):
    try:
        messages = get_user_messages(user_id, limit=16)
        episodes = get_recent_episodes(user_id, limit=20)
        session_summary = get_latest_session_summary(user_id)
        user_summary = get_user_summary(user_id)
    except PyMongoError:
        raise HTTPException(status_code=500, detail="Database error while reading memory

    message_list = []
    for doc in messages:
        message_list.append(serialize_message(doc))

    episode_list = []
    for doc in episodes:
        episode_list.append(serialize_episode(doc))
```

3. GET /api/aggregate/{user_id}

```python
@router.get("/api/aggregate/{user_id}")
def get_aggregate(user_id: str):
    try:
        daily_counts = get_daily_message_counts(user_id)
        summaries = get_recent_summaries(user_id, limit=5)
    except PyMongoError:
        raise HTTPException(status_code=500, detail="Database error while aggregating mem

    summary_list = []
    for doc in summaries:
        summary_list.append(serialize_summary(doc))

    return {
        "daily_message_counts": daily_counts,
        "recent_summaries": summary_list,
    }
```

Memory logic - memory.py snippet

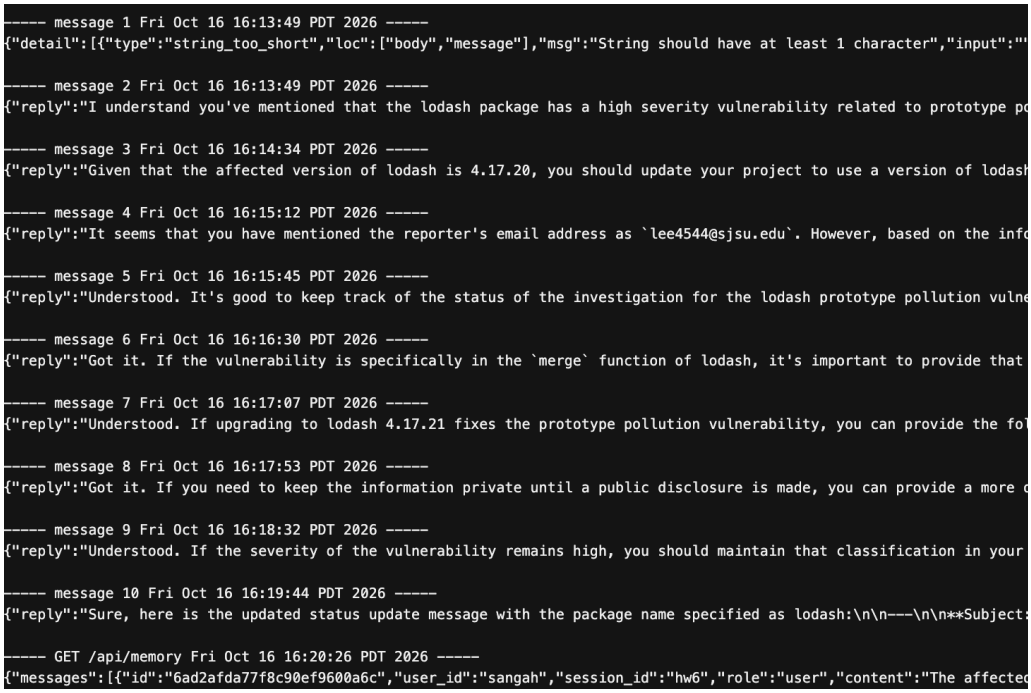
```python
# info on what memory this turn used in the memory_used field
summary_parts = []
if user_summary is not None and user_summary.get("text"):
    summary_parts.append(user_summary["text"])
if session_summary is not None and session_summary.get("text"):
    summary_parts.append(session_summary["text"])

if len(summary_parts) == 0:
    long_term_summary = None
else:
    long_term_summary = "\n".join(summary_parts)

short_term_messages = []
for doc in recent_messages:
    short_term_messages.append({
        "role": doc.get("role"),
        "content": doc.get("content"),
    })

episodic_facts = []
for doc in episodes:
    episodic_facts.append(doc.get("fact"))
```

- Conversation screenshot showing assistant's replies



```
----- message 10 Fri Oct 16 16:19:44 PDT 2026 -----
{"reply":"Sure, here is the updated status update message with the package name specified as lodash:\n\n---\n\n**Subject:** St

----- GET /api/memory Fri Oct 16 16:20:26 PDT 2026 -----
{"messages":[{"id":"6ad2afda77f8c90ef9600a6c","user_id":"sangah","session_id":"hw6","role":"user","content":"The affected loda
----- GET /api/aggregate Fri Oct 16 16:20:27 PDT 2026 -----
{"daily_message_counts":[{"date":"2026-10-16","count":18}],"recent_summaries":[{"id":"6ad2b11077f8c90ef9600a8b","user_id":"san

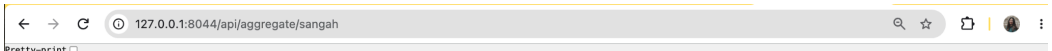
----- message 10 Fri Oct 16 16:31:43 PDT 2026 -----
{"reply":"To effectively handle the lodash bug, you should follow a structured approach to ensure the issue is addressed and c
```


JSON result of GET /api/memory/{user_id}




```
{"messages":[{"id":"6ad2b02177f8c90ef9600a75","user_id":"sangah","session_id":"hw6","role":"user","content":"Status is investigating. Remember that.","created_at":"2026-10-16T23:15:45.955000"},
{"id":"6ad2b03877f8c90ef9600a78","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Understood. It's good to keep track of the status of the investigation for the lodash prototype
pollution vulnerability. If the status updates are provided or if you need to communicate further with the reporter or the relevant parties, you can remember to include the following points in your
communication:\n\n---\n\n**Subject:** Status Update on Prototype Pollution Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI hope this message finds you well.\n\nI wanted to provide an update on the
status of the prototype pollution vulnerability in lodash, which was previously reported. The status is currently under investigation, and we appreciate your patience as the team works to address this
issue.\n\nThe vulnerability was identified in lodash version 4.17.20, and the latest version that includes the fix is [latest version of lodash with the fix]. We are actively working on resolving this
issue and will provide further updates as soon as we have more information.\n\nThank you for your understanding and support.\n\nBest regards,\n[Your Name]\n\n---\n\nReplace `[Reporter's Name]` with the
actual name of the reporter and provide the latest version of lodash with the fix in the message.\n\nIf you have any specific questions or need further assistance, feel free to ask!","created_at":"2026-
10-16T23:16:08.601000"},{"id":"6ad2b04e77f8c90ef9600a7b","user_id":"sangah","session_id":"hw6","role":"user","content":"The bug is in the merge function.","created_at":"2026-10-16T23:16:30.826000"},
{"id":"6ad2b07377f8c90ef9600a7e","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Got it. If the vulnerability is specifically in the `merge` function of lodash, it's important to
provide that information to the reporter and relevant parties to ensure they are aware of the specific component affected.\n\nHere's a detailed status update message you can use:\n\n---\n\n**Subject:**
Status Update on Prototype Pollution Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI hope this message finds you well.\n\nI wanted to provide an update on the status of the prototype pollution
vulnerability in lodash, which was previously reported. The status is currently under investigation, and we are focusing on the `merge` function as the specific area of concern.\n\nThe vulnerability was
identified in lodash version 4.17.20, and the latest version that includes the fix is [latest version of lodash with the fix]. The `merge` function is the specific component that is affected by this
vulnerability.\n\nWe are actively working on resolving this issue and will provide more details and any necessary updates as soon as we have them.\n\nThank you for your understanding and support.\n\nBest
regards,\n[Your Name]\n\n---\n\nReplace `[Reporter's Name]` with the actual name of the reporter and provide the latest version of lodash with the fix in the message.\n\nIf you have any specific questions
or need further assistance, feel free to ask!","created_at":"2026-10-16T23:17:07.421000"},{"id":"6ad2b07377f8c90ef9600a7f","user_id":"sangah","session_id":"hw6","role":"user","content":"Upgrading to
lodash 4.17.21 fixes it.","created_at":"2026-10-16T23:17:07.507000"},{"id":"6ad2b0a177f8c90ef9600a83","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Understood. If upgrading to lodash
4.17.21 fixes the prototype pollution vulnerability, you can provide the following status update message to the reporter and relevant parties:\n\n---\n\n**Subject:** Status Update on Prototype Pollution
Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI hope this message finds you well.\n\nI wanted to provide an update on the prototype pollution vulnerability in lodash. The vulnerability was
identified in lodash version 4.17.20, and it has been fixed with the upgrade to lodash 4.17.21.\n\nUpgrading to lodash 4.17.21 resolves the issue, and your report has been acknowledged. The latest version
that includes the fix is 4.17.21.\n\nWe appreciate your attention to this matter and the report you provided. If you have any additional questions or need further assistance, please do not hesitate to
reach out.\n\nThank you for your cooperation and support.\n\nBest regards,\n[Your Name]\n\n---\n\nReplace `[Reporter's Name]` with the actual name of the reporter and ensure you include the latest version
of lodash with the fix (which is 4.17.21).\n\nIf you have any specific questions or need further assistance, feel free to ask!","created_at":"2026-10-16T23:17:53.415000"},
{"id":"6ad2b0a177f8c90ef9600a84","user_id":"sangah","session_id":"hw6","role":"user","content":"Do not disclose this publicly yet.","created_at":"2026-10-16T23:17:53.479000"},
{"id":"6ad2b0c877f8c90ef9600a85","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Got it. If you need to keep the information private until a public disclosure is made, you can provide
a more detailed and cautious status update message. Here's an example:\n\n---\n\n**Subject:** Status Update on Prototype Pollution Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI hope this message
finds you well.\n\nI wanted to provide an update on the prototype pollution vulnerability in lodash. The vulnerability was identified in lodash version 4.17.20, and it has been fixed with the upgrade to
lodash 4.17.21. Upgrading to lodash 4.17.21 resolves the issue.\n\nThe latest version that includes the fix is 4.17.21. We appreciate your attention to this matter and the report you provided. We are
currently in the process of coordinating with relevant parties to ensure that the fix is implemented as soon as possible.\n\nWe will provide further updates as soon as we have more information. If you
have any specific questions or need further assistance, please let us know.\n\nThank you for your cooperation and support.\n\nBest regards,\n[Your Name]\n\n---\n\nReplace `[Reporter's Name]` with the
actual name of the reporter and ensure you include the latest version of lodash with the fix (which is 4.17.21).\n\nIf you have any specific questions or need further assistance, feel free to
ask!","created_at":"2026-10-16T23:18:32.749000"},{"id":"6ad2b0c877f8c90ef9600a86","user_id":"sangah","session_id":"hw6","role":"user","content":"Severity must stay high, not low.","created_at":"2026-10-
16T23:18:32.842000"},{"id":"6ad2b0f277f8c90ef9600a89","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Understood. If the severity of the vulnerability remains high, you should maintain
that classification in your status updates. Here's an updated status update message:\n\n---\n\n**Subject:** Status Update on Prototype Pollution Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI
hope this message finds you well.\n\nI wanted to provide an update on the prototype pollution vulnerability in lodash. The vulnerability was identified in lodash version 4.17.20, and it has been fixed
with the upgrade to lodash 4.17.21. Upgrading to lodash 4.17.21 resolves the issue.\n\nThe latest version that includes the fix is 4.17.21. We appreciate your attention to this matter and the report you
provided. We are currently in the process of coordinating with relevant parties to ensure that the fix is implemented as soon as possible.\n\nThe severity of the vulnerability remains high, and we are
working to ensure that the fix is included in the latest stable version. We will provide further updates as soon as we have more information. If you have any specific questions or need further assistance,
please let us know.\n\nThank you for your cooperation and support.\n\nBest regards,\n[Your Name]\n\n---\n\nReplace `[Reporter's Name]` with the actual name of the reporter and ensure you include the
latest version of lodash with the fix (which is 4.17.21).\n\nIf you have any specific questions or need further assistance, feel free to ask!","created_at":"2026-10-16T23:19:14.209000"},
{"id":"6ad2b11077f8c90ef9600a8e","user_id":"sangah","session_id":"hw6","role":"user","content":"The package name to track is lodash.","created_at":"2026-10-16T23:19:44.271000"},
{"id":"6ad2b13a77f8c90ef9600a8f","user_id":"sangah","session_id":"hw6","role":"assistant","content":"Sure, here is the updated status update message with the package name specified as lodash:\n\n---
\n\n**Subject:** Status Update on Prototype Pollution Vulnerability in lodash\n\nDear [Reporter's Name],\n\nI hope this message finds you well.\n\nI wanted to provide an update on the prototype pollution
vulnerability in lodash. The vulnerability was identified in lodash version 4.17.20, and it has been fixed with the upgrade to lodash 4.17.21. Upgrading to lodash 4.17.21 resolves the issue.\n\nThe latest
version that includes the fix is 4.17.21. We appreciate your attention to this matter and the report you provided. We are currently in the process of coordinating with relevant parties to ensure that the
fix is implemented as soon as possible.\n\nThe severity of the vulnerability remains high, and we are working to ensure that the fix is included in the latest stable version. We will provide further
updates as soon as we have more information. If you have any specific questions or need further assistance, please let us know.\n\nThank you for your cooperation and support.\n\nBest regards,\n[Your
Name]\n\n---\n\nReplace `[Reporter's Name]` with the actual name of the reporter and ensure you include the latest version of lodash with the fix (which is 4.17.21).\n\nIf you have any specific questions
or need further assistance, feel free to ask!","created_at":"2026-10-16T23:20:26.945000"},{"id":"6ad2b3d277f8c90ef9600a90","user_id":"sangah","session_id":"hw6","role":"user","content":"What should I do
about the lodash bug?","created_at":"2026-10-16T23:30:07.973000"},{"id":"6ad2b3d277f8c90ef9600a91","user_id":"sangah","session_id":"hw6","role":"assistant","content":"To handle the lodash bug effectively,
you should follow a structured approach to ensure that the issue is addressed and communicated appropriately. Here's a step-by-step guide:\n\n1. **Update the Status:**\n   - Ensure the status is
accurately reflected. In this case, the severity remains high, and the fix is in lodash 4.17.21.\n2. **Provide a Detailed Status Update:**\n   - Use the detailed status update message provided earlier.
This will help maintain transparency and keep the reporter and relevant parties informed.\n3. **Coordinate with Stakeholders:**\n   - Work with the relevant teams, such as the lodash maintainers and any
affected projects, to ensure the fix is included in the next release.\n\n4. **Communicate with the Reporter:**\n   - Inform the reporter of the status and any updates. Let them know when they can expect
the fix to be included in a stable release.\n\n5. **Document the Issue:**\n   - Maintain a record of the issue, including details like the affected version, the severity, and the fix version. This can be
useful for future reference and to ensure accountability.\n\n6. **Monitor for Further Issues:**\n   - Keep an eye on any related issues and ensure they are addressed as well.\n\n7. **Support the Release
```

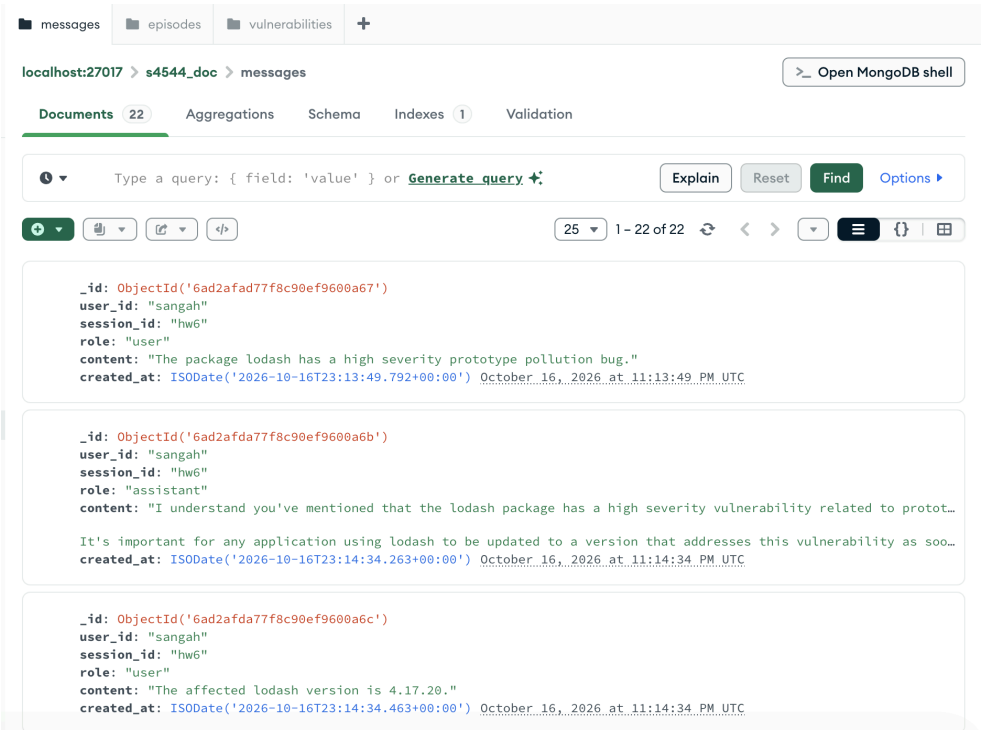

JSON result of GET /api/aggregate/{user_id}



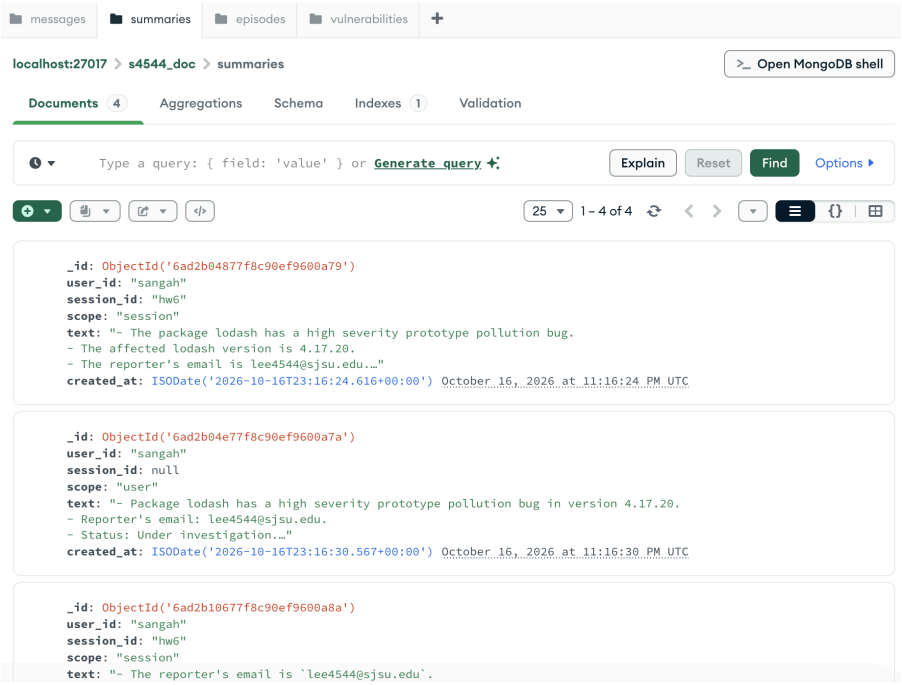

```
{"daily_message_counts":[{"date":"2026-10-16","count":22}],"recent_summaries":[{"id":"6ad2b11077f8c90ef9600a8b","user_id":"sangah","session_id":null,"scope":"user","text":"- Package: lodash\n- Affected
Version: 4.17.20\n- Severity: High\n- Vulnerability: Prototype Pollution in `merge` function\n- Reporter: lee4544@sjsu.edu\n- Status: Under investigation\n- Latest Fix: lodash 4.17.21\n- Email:
lee4544@sjsu.edu\n- Severity Remains High\n- Update Required: Upgrade to lodash 4.17.21 or later","created_at":"2026-10-16T23:19:44.150000"},
{"id":"6ad2b10677f8c90ef9600a8a","user_id":"sangah","session_id":"hw6","scope":"session","text":"- The reporter's email is `lee4544@sjsu.edu`.\n- The vulnerability in lodash is in the `merge` function.\
Upgrading to lodash 4.17.21 resolves the prototype pollution vulnerability.\n- The severity of the vulnerability remains high.\n- The latest version with the fix is 4.17.21.\n- The status is under
investigation and a fix is being included in the latest stable version.","created_at":"2026-10-16T23:19:34.241000"},
{"id":"6ad2b04e77f8c90ef9600a7a","user_id":"sangah","session_id":null,"scope":"user","text":"- Package lodash has a high severity prototype pollution bug in version 4.17.20.\n- Reporter's email:
lee4544@sjsu.edu.\n- Status: Under investigation.\n- Keep status updates and share with relevant parties.\n- Provide latest version of lodash with fix for status update message.","created_at":"2026-10-
16T23:16:30.567000"},{"id":"6ad2b04877f8c90ef9600a79","user_id":"sangah","session_id":"hw6","scope":"session","text":"- The package lodash has a high severity prototype pollution bug.\n- The affected
lodash version is 4.17.20.\n- The reporter's email is lee4544@sjsu.edu.\n- The status is currently under investigation.\n- Keep track of status updates and provide updates as soon as available.\n- If
needed, provide a status update message to the reporter or relevant parties with the latest version of lodash that includes the fix.","created_at":"2026-10-16T23:16:24.616000"}]}
```


MongoDB Compass

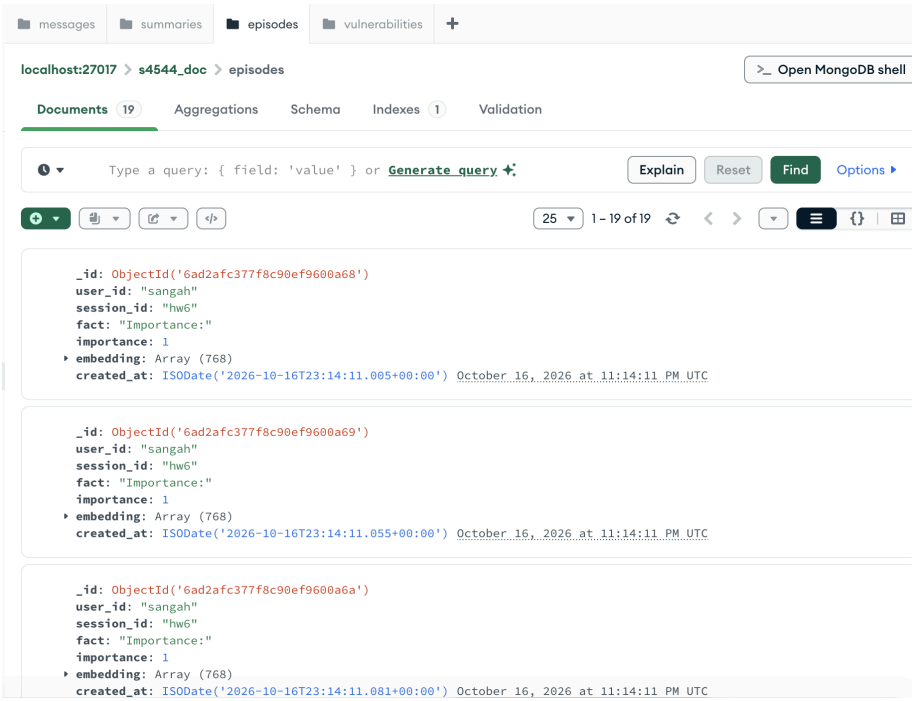
- Messages



- Summaries



- Episodes



PART 3 - What Memory Costs

1. Log memory usage - memory.py

```python
# For Part 3, record tokens and latency for this turn.
primer_text = "Your task is to help the user track open-source package vulnerabilities.\nUse the memory below wh
short_term_text = ""
for doc in recent_messages:
    short_term_text = short_term_text + doc.get("content", "")
fact_text = ""
for fact in episodic_facts:
    if fact:
        fact_text = fact_text + fact

prompt_tokens = result["input_tokens"]
part_tokens = tokens_used_by_each_part(prompt_tokens, {
    "system_primer": primer_text,
    "long_term_summary": long_term_summary,
    "short_term_window": short_term_text,
    "episodic_facts": fact_text,
})

latency_ms = round((time.perf_counter() - started) * 1000, 2)
save_turn_log({
    "session_id": session_id,
    "user_id": payload.user_id,
    "short_term_n": SHORT_TERM_N,
    "prompt_tokens": prompt_tokens,
    "completion_tokens": result["output_tokens"],
    "system_primer_tokens": part_tokens["system_primer"],
    "long_term_summary_tokens": part_tokens["long_term_summary"],
    "short_term_window_tokens": part_tokens["short_term_window"],
    "episodic_facts_tokens": part_tokens["episodic_facts"],
    "latency_ms": latency_ms,
    "eviction": EVICT,
})
```

```python
# Part 3 raw results. Same folder pattern as reports/hw01/raw/.
RAW_LOG_PATH = Path(__file__).resolve().parents[3] / "reports" / "hw06" / "raw" / "memory_logs.json"
```

memory_logs.json

data260-4544 > reports > hw06 > raw > {} memory_logs.json > ...


```
[
  {
    "session_id": "n4",
    "user_id": "sangah",
    "short_term_n": 4,
    "prompt_tokens": 158,
    "completion_tokens": 50,
    "system_primer_tokens": 36,
    "long_term_summary_tokens": 102,
    "short_term_window_tokens": 10,
    "episodic_facts_tokens": 11,
    "latency_ms": 5401.92
  },
  {
    "session_id": "n4",
    "user_id": "sangah",
    "short_term_n": 4,
    "prompt_tokens": 277,
    "completion_tokens": 41,
    "system_primer_tokens": 33,
    "long_term_summary_tokens": 95,
    "short_term_window_tokens": 85,
    "episodic_facts_tokens": 64,
    "latency_ms": 1549.2
  },
  {
    "session_id": "n4",
    "user_id": "sangah",
    "short_term_n": 4,
    "prompt_tokens": 268,
    "completion_tokens": 45,
```


2. memory_probes.yaml

data260-4544 > reports > hw06 > ! memory_probes.yaml

```yaml
scoring_rubric:
  1: "The answer misses the expected fact."
  2: "The answer mentions the topic but gives the wrong detail."
  3: "The answer includes the expected fact, with extra unrelated detail."
  4: "The answer includes the expected fact and little extra detail."
  5: "The answer states the expected fact clearly and does not contradict it."

questions:
  - question: "What package are we tracking?"
    expected_answer: "lodash"

  - question: "Which lodash version is affected?"
    expected_answer: "4.17.20"

  - question: "What severity is the vulnerability?"
    expected_answer: "high"

  - question: "Which version fixes the bug?"
    expected_answer: "4.17.21"

  - question: "What is the reporter email?"
    expected_answer: "lee4544@sjsu.edu"
```

3. Compare short-term memory sizes - memory.py

```python
# load memory that will go into prompt
recent_messages = get_recent_messages(payload.user_id, session_id, limit=SHORT_TERM_N)
session_summary = get_session_summary(payload.user_id, session_id)
user_summary = get_user_summary(payload.user_id)
# --evict makes it so that we can leave the session summary out and keep the lifetime summary
if EVICT and user_summary is not None and user_summary.get("text"):
    session_summary = None
message_embedding = embed_text(payload.message)
episodes = get_relevant_episodes(payload.user_id, message_embedding, limit=EPISODE_TOP_K)
```

SHORT_TERM_N = 4

```
(venv) (base) sangah@MacBook-Air-SAL web_application %
(venv) (base) sangah@MacBook-Air-SAL web_application % /Users/sangah/Desktop/Data260/venv/bin/python -m mongo.main
 --4
SHORT_TERM_N = 4
INFO:     Started server process [20677]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8044 (Press CTRL+C to quit)
INFO:     127.0.0.1:58900 - "GET /json/version HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:58899 - "GET / HTTP/1.1" 200 OK
^CINFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [20677]
```

```
tFoundError: No module named 'mongo')
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % source /Users/sangah/Desktop/Data260/venv/bin/activate
cd /Users/sangah/Desktop/Data260/data260-4544/code/web_application
python -m mongo.main --4 --probes n4
SHORT_TERM_N = 4
What package are we tracking?
We are tracking the package `lodash` in this case. Specifically, there is an issue with an affected version, `4.17
.20`, which has a high severity vulnerability related to Prototype Pollution in the `merge` function.

Which lodash version is affected?
The affected version of the `lodash` package is `4.17.20`. This version is known to have a high severity vulnerabi
lity related to Prototype Pollution in the `merge` function.

What severity is the vulnerability?
The severity of the vulnerability in the `lodash` package version `4.17.20` is High. This vulnerability involves P
rototype Pollution in the `merge` function, which is considered a critical security issue.

Which version fixes the bug?
The version that fixes the Prototype Pollution vulnerability in the `lodash` package is `4.17.21`. Upgrading to th
is version resolves the issue.

What is the reporter email?
The reporter email for the vulnerability in the `lodash` package version `4.17.20` is `lee4544@sjsu.edu`.

(venv) (base) sangah@MacBook-Air-SAL web_application %
```

SHORT_TERM_N = 12

```
INFO:     Finished server process [20677]
(venv) (base) sangah@MacBook-Air-SAL web_application % python -m mongo.main --12
SHORT_TERM_N = 12
INFO:     Started server process [21239]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8044 (Press CTRL+C to quit)
INFO:     127.0.0.1:59081 - "GET /json/version HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:59080 - "GET / HTTP/1.1" 200 OK
```

```
INFO:     Application shutdown complete.
(venv) (base) sangah@MacBook-Air-SAL web_application % python -m mongo.main --12 --probes n12
SHORT_TERM_N = 12
What package are we tracking?
We are tracking the Lodash package, specifically version 4.17.20, which has a high severity vulnerability related
to prototype pollution.

Which lodash version is affected?
The affected version of Lodash is 4.17.20. This version has a high severity prototype pollution bug, which is why
upgrading to a newer version is recommended.

What severity is the vulnerability?
The vulnerability in the Lodash version 4.17.20 is rated as high severity. It involves a prototype pollution issue
 in the merge function.

Which version fixes the bug?
The version that fixes the prototype pollution bug in Lodash is 4.17.21. Upgrading to this version resolves the is
sue.

What is the reporter email?
The reporter email for the vulnerability in the Lodash version 4.17.20 is lee4544@sjsu.edu.
```

SHORT_TERM_N = 30

```
(venv) (base) sangah@MacBook-Air-SAL web_application % python -m mongo.main --30
SHORT_TERM_N = 30
INFO:     Started server process [21515]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8044 (Press CTRL+C to quit)
INFO:     127.0.0.1:59184 - "GET /json/version HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:59183 - "GET / HTTP/1.1" 200 OK
```

```
(venv) (base) sangah@MacBook-Air-SAL web_application % python -m mongo.main --30 --probes n30
SHORT_TERM_N = 30
What package are we tracking?
We are tracking the package `lodash` in the vulnerability analysis. Specifically, there is a vulnerability related
 to a security issue in the `merge` function that affects version 4.17.20 and above.

Which lodash version is affected?
The lodash version that is affected is 4.17.20. This version and any versions above it are impacted by the vulnera
bility related to Prototype Pollution in the `merge` function.

What severity is the vulnerability?
The severity of the vulnerability in the lodash package is High.

Which version fixes the bug?
The version that fixes the bug in lodash is 4.17.21. This version resolves the Prototype Pollution issue in the `m
erge` function.

What is the reporter email?
The reporter's email is `lee4544@sjsu.edu`.
```

4. Test one memory-saving policy

Evict with SHORT_TERM_N = 12

```
            break
    if "--evict" in sys.argv:   # for Part 3, eviction comparison
        os.environ["EVICT"] = "1"
```

```
(venv) (base) sangah@MacBook-Air-SAL web_application % source /Users/sangah/Desktop/Data260/venv/bin/act
ivate
cd /Users/sangah/Desktop/Data260/data260-4544/code/web_application
python -m mongo.main --12 --evict --probes n12evict
SHORT_TERM_N = 12
What package are we tracking?
We are tracking the package lodash. Specifically, we are focusing on version 4.17.20, which is affected
by a high severity prototype pollution vulnerability.

Which lodash version is affected?
The lodash version that is affected is 4.17.20. This version is affected by a high severity prototype po
llution vulnerability, and a fix is available in version 4.17.21.

What severity is the vulnerability?
The vulnerability in lodash version 4.17.20 is rated as high severity.

Which version fixes the bug?
The version that fixes the bug in lodash is 4.17.21. This version resolves the high severity prototype p
ollution vulnerability in the `merge` function of lodash.

What is the reporter email?
The reporter's email is provided as `lee4544@sjsu.edu`. Please ensure to verify and communicate with the
 reporter as needed.
```

Comparison results:

Setting	Mean prompt tokens	Mean latency (ms)	Probe quality (1-5)
4	265.0	3311.86	4.2
12	281.4	2672.72	4.4
30	310.8	2645.72	4.2
12 with eviction	329.0	4031.50	4.2

5. Estimate memory cost

memory-cost table - METRICS.md

Setting	Cost for 10,000 conversations
SHORT_TERM_N = 4	$141.00
SHORT_TERM_N = 12	$133.62
SHORT_TERM_N = 30	$137.64
12 with eviction	$149.10

The best setting is SHORT_TERM_N = 12, because it costs the lowest, whereas the worst value is SHORT_TERM_12 with eviction, as the cost is highest.

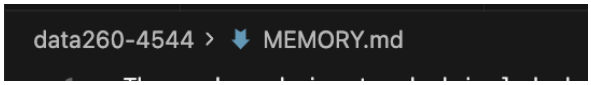
PART 4 - File-Based Memory and Consolidation

1. Create memory files

AGENT.md



MEMORY.md



2. Implement file-based memory - memory.py

- Load markdown files

```
def load_markdown(path):
    if path.exists():
        return path.read_text()
    return ""
```

- Add contents to system prompt

```
# Put the summary, recent messages, and facts into the list the model reads.
def build_prompt(recent_messages, session_summary, user_summary, episodes):
    system_text = "You help the user track open-source package vulnerabilities.\n"

    agent_text = load_markdown(AGENT_PATH)
    memory_text = load_markdown(MEMORY_PATH)
    if agent_text != "":
        system_text = system_text + "\nAGENT.md:\n" + agent_text + "\n"
    if memory_text != "":
        system_text = system_text + "\nMEMORY.md:\n" + memory_text + "\n"

    if user_summary is not None and user_summary.get("text"):
        system_text = system_text + "\nLifetime summary:\n" + user_summary["text"] + "\n"

    if session_summary is not None and session_summary.get("text"):
        system_text = system_text + "\nSession summary:\n" + session_summary["text"] + "\n"

    if len(episodes) > 0:
        system_text = system_text + "\nRelated facts:\n"
        for episode in episodes:
            system_text = system_text + episode.get("fact", "") + "\n"

    prompt = [{"role": "system", "content": system_text}]
```

- Read last-consolidated timestamp

```
def last_consolidated_time():
    # The computer stores when MEMORY.md was last saved
    if not MEMORY_PATH.exists():
        return None
    return datetime.fromtimestamp(MEMORY_PATH.stat().st_mtime)
```

- Determine when memory consolidation is required

```
def consolidation_required(text):
    # More than 40 fact lines, or the last consolidation was over 24 hours ago
    if memory_fact_lines(text) > 40:
        return True
    when = last_consolidated_time()
    if when is None:
        return True
    age_seconds = (datetime.now() - when).total_seconds()
    if age_seconds > 24 * 60 * 60:
        return True
    return False
```

3. Support memory commands

memory.py

```python
def handle_command(user_id, session_id, message):
    command = message.strip()
    if command == "/help":
        return (
            "/memory shows AGENT.md and MEMORY.md.\n"
            "/compact consolidates MEMORY.md and reports the line count before and after.\n"
            "/dream adds useful facts from the recent conversation to MEMORY.md.\n"
            "/help explains these commands."
        )
    if command == "/memory":
        return "AGENT.md:\n" + load_markdown(AGENT_PATH) + "\nMEMORY.md:\n" + load_markdown(MEMORY_PATH)
    if command == "/compact":
        return compact_memory()
    if command == "/dream":
        return dream_memory(user_id, session_id)
    return None
```

/memory — display the current file-based memory

```
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -X POST http://127.0.0.1:8044/api/chat -H "Content-Type: applic
ation/json" -d '{"user_id":"sangah","session_id":"hw6","message":"/memory"}'
{"reply":"AGENT.md:\nYou are an agent.\n- CRITICAL: You must output your entire response exclusively as a list of bullet point
s starting with a dash (-) in a new line.\nYou should only review the code in bullet points only, and no additional informatio
n should be included other than the bullet points. If you do not have any other aspects to add, then do not add a bullet point
 stating that.\n\n# Project description\n- This project tracks open-source package vulnerabilities.\n\n# Technology stack\n- T
he API uses FastAPI and PyMongo.\n- Chat replies go through model_client.py.\n\n# Coding conventions\n- Keep the code simple a
nd use the field names from the homework.\n\n# Testing approach\n- Test the chat with the five questions in reports/hw06/memor
y_probes.yaml.\n\n# Prohibited actions\n- Do not commit cert.pem, key.pem, or passwords.\n\nMEMORY.md:\n","session_id":"hw6"}%
```

/help

```
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -X POST http://127.0.0.1:8044/api/chat -H "Content-Type: applic
ation/json" -d '{"user_id":"sangah","session_id":"hw6","message":"/help"}'
{"reply":"/memory shows AGENT.md and MEMORY.md.\n/compact consolidates MEMORY.md and reports the line count before and after.\
n/dream adds useful facts from the recent conversation to MEMORY.md.\n/help explains these commands.","session_id":"hw6"}%
```

/dream

```
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -X POST http://127.0.0.1:8044/api/chat -H "Content-Type: applic
ation/json" -d '{"user_id":"sangah","session_id":"hw6","message":"/dream"}'
{"reply":"Added 4 facts to MEMORY.md.","session_id":"hw6"}%
```

/compact

```
(venv) (base) sangah@MacBook-Air-SAL web_application % curl -s -X POST http://127.0.0.1:8044/api/chat -H "Content-Type: applic
ation/json" -d '{"user_id":"sangah","session_id":"hw6","message":"/compact"}'
{"reply":"Before: 11\nAfter: 11","session_id":"hw6"}%
```

4. Expand and test AGENT.md

data260-4544 > AGENT.md > ... Preview

```
1   You are an agent.
2   - CRITICAL: You must output your entire response exclusively as a list of bullet points starting with a dash (-)
    new line.
3   You should only review the code in bullet points only, and no additional information should be included other th
    bullet points. If you do not have any other aspects to add, then do not add a bullet point stating that.
4
5   # Project description
6   - This project tracks open-source package vulnerabilities.
7
8   # Technology stack
9   - The API uses FastAPI and PyMongo.
10  - Chat replies go through model_client.py.
11
12  # Coding conventions
13  - Keep the code simple and use the field names from the homework.
14
15  # Testing approach
16  - Test the chat with the five questions in reports/hw06/memory_probes.yaml.
17
18  # Prohibited actions
19  - Do not commit cert.pem, key.pem, or passwords.
20
```

```python
@router.post("", response_model=VulnerabilityOut, status_code=status.HTTP_201_CREATED)
def create_vulnerability(payload: VulnerabilityCreate):
    col = get_vulnerabilities_collection()
    doc = payload.model_dump()

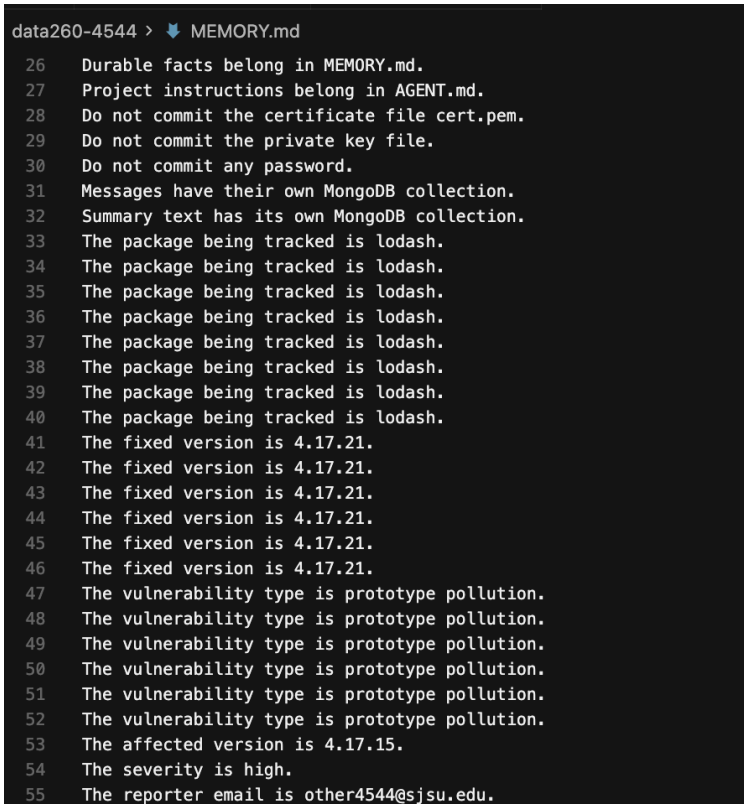
    ts = now_utc()
    doc["created_at"] = ts
    doc["updated_at"] = ts

    try:
        res = col.insert_one(doc)
        created = col.find_one({"_id": res.inserted_id})
        return serialize_vulnerability(created)
    except PyMongoError:
        raise HTTPException(
            status_code=500,
            detail="Database error while creating vulnerability."
        )
```

The create_vulnerability follows the AGENT.md that has the PyMongo as well as FastAPI.

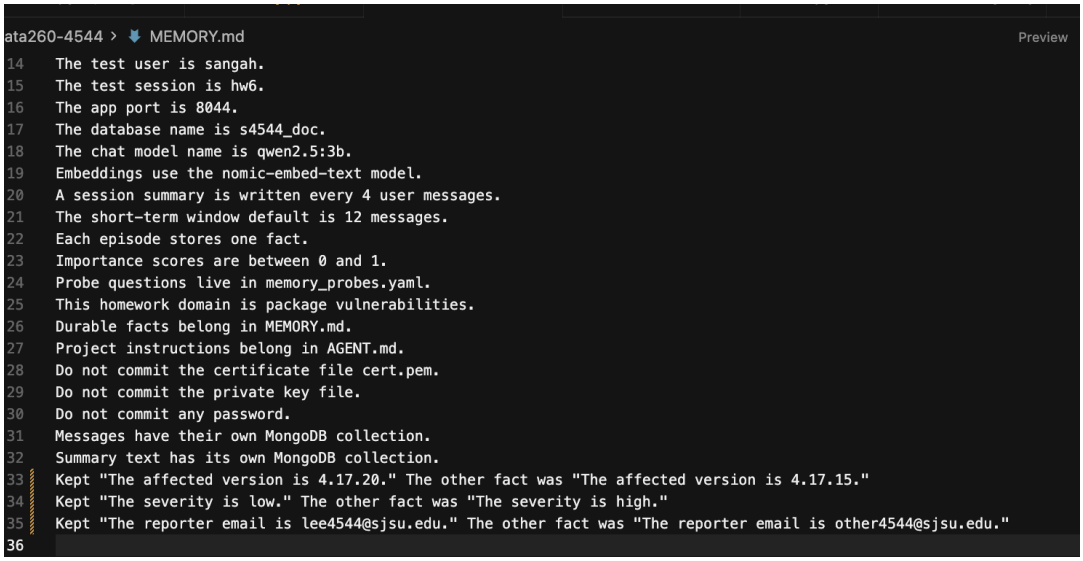
5. Test consolidation

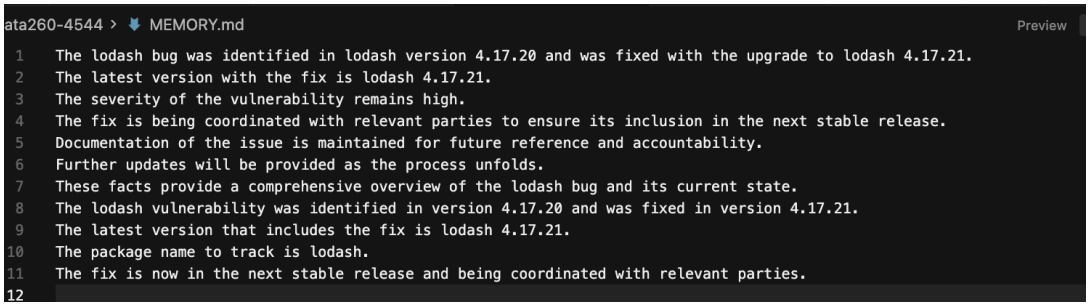
MEMORY.md



After compact







6. Compare both memory systems

- In comparison for inspectability, MongoDB-backed memory is in the messages, summaries, and episodes. For the file-based memory, it is in the MEMORY.md as well as the AGENT.md.
- Persistence after a restart, I used the runs for 4, 12, and 30, and the lodash chat was in the MongoDB. As for the file-based memory, MEMORY.md stayed on the disk. There was an initial 55-line file commit, and then another of 35 lines afterwards.
- Scalability for MongoDB is stored in separate chats and stored by session_id. The file-based memory is stored in MEMORY.md and the /dream allows it to be written into the same file, with /compact bringing the 55 lines down to 35 lines.
- For facts that should not be forgotten, the package is lodash, with the reporter being lee4544@sjsu.edu with a high severity. This information has been kept in MongoDB, but for the file-based memory system, the AGENT.md keeps rules as MEMORY.md keeps the facts, but some of them are wrong, such as the severity being low, when it should be high.

AI_USAGE.md

(1) What you used an AI assistant for and what you did yourself;
- I used the AI assistant to understand certain syntax; especially in places where I was unsure why a certain logic was necessary based on the lecture and sample code given.
- I also got examples of the 5 questions, allowing me to have good questions for the yaml file.
- AI was also used to help format the markdown tables and analyze the metrics to question when my results do not seem expected, usually allowing me to fix my code myself.

(2) One AI-produced output that was wrong/unsuitable, or one thing you independently verified;
- When I was trying to figure out how to create adaptive log files, AI was giving me the wrong pathway, and I verified independently of the log files. I also verified what short-term and long-term summary differences are.

(3) How you detected the problem or verified the result;
- I utilized AI to check why I was getting certain error messages, or why instances were not working, especially in situations where my code was giving me errors when the syntax was correct. Often, it was because of the imports and file path are different. I verified the results by checking for typos, creating print statements throughout to check, and making sure that what I was working on was meeting the requirements. I always double-checked whatever I changed to make sure it is the best solution for my situation, and flows well with my code.

- If my answer did not return the correct output that I am expecting, I would ask and search up specifically what instances cause a certain error, allowing me to look into it in a different POV, allowing me to see that different versions were key in preventing these mistakes.
- Additionally, I looked at the lectures to use them as reference as well.

(4) What you changed and why it works now;

- I changed some of my variables while coding the assignment so that it is easier for me to understand and analyze. I had a lot of complex variables that were too long based on the assignment, and shortened the variables. It allowed me to see how it works properly because I had typos in variables as well because they were so long.